

TONGYEONG, Korea, Republic of

WMO: 471620

Lat: 34.8456N

Lon: 128.4356E

Elev: 33

StdP: 100.93

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.5	(c) -2.9	(d) -18.4	(e) 0.7	(f) -0.6	(g) -16.5	(h) 0.9	(i) -0.1	(j) 8.1	(k) 1.5	(l) 7.3	(m) 2.4	(n) 3.1	(o) 0	(p) 0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.1	(c) 31.0	(d) 25.2	(e) 29.8	(f) 24.9	(g) 28.8	(h) 24.6	(i) 26.5	(j) 29.2	(k) 26.0	(l) 28.4	(m) 25.5	(n) 27.9	(o) 3.1	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 25.7	(b) 21.1	(c) 28.0	(d) 25.2	(e) 20.4	(f) 27.6	(g) 24.7	(h) 19.8	(i) 27.2	(j) 83.0	(k) 29.2	(l) 80.8	(m) 28.5	(n) 78.6	(o) 27.8	(p) 28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
8.2	6.5	5.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-7.2	33.1	1.7	1.4	-8.4	34.2	-9.4	35.0	-10.4	35.8	-11.6	36.8
			-9.4	26.8	1.5	0.9	-10.4	27.5	-11.3	28.0	-12.1	28.5	-13.2	29.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	3.3	4.9	8.7	13.4	17.7	21.1	24.7	26.3	22.7	17.9	11.6	5.6
	DBStd	8.18	3.04	3.32	3.03	2.56	2.01	1.53	2.19	1.94	2.15	2.70	3.50	3.45
	HDD10.0	582	208	144	61	4	0	0	0	0	0	0	24	141
	HDD18.3	1962	466	375	300	150	35	1	0	0	0	40	202	394
	CDD10.0	2367	1	2	20	105	240	332	457	507	382	244	72	5
	CDD18.3	706	0	0	0	1	17	83	198	248	133	26	1	0
	CDH23.3	4379	0	0	0	1	35	152	1308	2229	611	44	0	0
	CDH26.7	938	0	0	0	0	1	5	255	600	77	1	0	0
Wind	WSAvg	2.5	2.7	2.7	2.8	2.6	2.3	2.1	2.5	2.5	2.4	2.3	2.3	2.5
Precipitation	PrecAvg	1423	32	49	81	144	146	201	252	234	155	54	51	28
	PrecMax	2079	110	125	191	327	402	558	494	606	450	196	127	72
	PrecMin	845	1	0	12	35	14	24	91	46	15	0	2	1
	PrecStd	328	28	32	41	74	95	116	116	149	102	50	39	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	15.6	17.6	22.0	26.1	27.3	32.4	33.1	30.0	25.9	21.2	16.1
		MCWB	10.3	12.2	11.9	13.9	17.1	19.9	25.4	26.2	23.9	19.3	16.1	12.0
	2%	DB	11.0	13.5	16.2	20.1	24.2	25.9	30.4	31.6	28.5	24.5	19.5	14.1
		MCWB	6.6	9.6	11.8	13.7	16.5	19.6	25.3	25.5	22.8	18.7	14.8	10.2
	5%	DB	9.4	11.9	15.0	18.8	22.8	24.9	29.1	30.5	27.5	23.4	18.2	12.3
		MCWB	5.3	7.7	10.8	13.6	16.3	19.9	25.0	25.1	22.4	18.1	13.8	8.3
	10%	DB	8.0	10.3	13.8	17.7	21.6	24.1	28.0	29.5	26.5	22.3	16.9	11.0
		MCWB	4.3	6.3	9.7	13.1	16.3	20.1	24.6	24.9	22.0	17.2	13.0	7.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	13.9	15.0	17.9	20.2	23.6	26.9	27.3	25.7	22.4	18.0	13.5
		MCDB	12.9	14.9	15.8	19.0	21.6	24.4	29.7	30.5	28.0	24.0	19.6	14.9
	2%	WB	7.7	10.8	13.5	16.4	19.0	22.7	26.2	26.7	24.9	21.0	16.5	10.8
		MCDB	9.8	12.4	14.9	17.9	21.2	23.7	28.6	29.5	27.2	23.2	18.3	13.0
	5%	WB	6.3	8.4	11.9	15.2	18.2	22.0	25.7	26.3	24.0	19.5	15.0	9.2
		MCDB	8.5	11.0	14.1	17.3	20.6	23.2	28.0	28.9	26.3	21.9	17.3	11.6
	10%	WB	4.9	6.8	10.3	14.2	17.5	21.2	25.2	25.8	23.1	18.2	13.8	7.7
		MCDB	7.6	9.5	13.0	16.5	20.1	22.9	27.4	28.3	25.4	21.1	16.3	10.5
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.6	7.4	7.2	6.7	5.5	4.5	5.1	6.1	7.4	7.6	7.6
		MCDBR	8.2	8.7	8.2	8.3	8.6	7.1	6.0	6.5	7.0	7.9	8.0	8.4
		MCWBR	5.6	5.9	5.2	4.4	3.8	3.2	2.3	2.4	3.1	3.7	4.7	5.8
	5% WB	MCDBR	6.9	7.7	6.9	6.0	6.0	4.1	5.2	5.1	5.3	6.1	6.7	7.4
		MCWBR	5.8	6.5	6.1	4.9	3.8	2.8	2.4	2.3	3.1	3.9	5.0	6.0
Clear-Sky Solar Irradiance	taub	0.414	0.463	0.535	0.562	0.576	0.617	0.589	0.546	0.497	0.459	0.429	0.407	
	taud	2.073	1.979	1.822	1.810	1.831	1.814	1.942	2.058	2.137	2.118	2.115	2.105	
	Ebn at Noon	753	759	737	744	740	708	725	749	767	757	733	733	
	Edh at Noon	128	157	200	212	210	214	187	164	145	135	122	118	
All-Sky Solar Radiation	RadAvg	2.71	3.38	4.39	5.04	5.53	4.87	4.69	4.77	4.05	3.72	2.80	2.47	
	RadStd	0.21	0.34	0.37	0.39	0.61	0.43	0.69	0.58	0.42	0.34	0.31	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+58	N/A

Nomenclature: See separate page